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WAVEGUIDES AND METHODS OF FORMING THE SAME

Abstract

In an embodiment, a method includes: depositing a waveguide cladding layer over a substrate, the substrate having a center portion in a cross-sectional view and an edge portion adjacent the center portion in the cross-sectional view; reshaping the waveguide cladding layer to form a bevel profile in the waveguide cladding layer over the edge portion of the substrate; forming a recess in the waveguide cladding layer; depositing a waveguide core layer in the recess and over the waveguide cladding layer, the waveguide core layer extending along the bevel profile of the waveguide cladding layer; and planarizing the waveguide core layer and the waveguide cladding layer to form a waveguide core, the waveguide core including a portion of the waveguide core layer in the recess.

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Background/Summary

BACKGROUND

[0001] Electrical signaling and processing are one technique for signal transmission and processing. Optical signaling and processing have been used in increasingly more applications in recent years, particularly due to the use of optical fiber-related applications for signal transmission. Optical signaling and processing are typically combined with electrical signaling and processing to provide full-fledged applications. For example, waveguides may be used for optical signal transmission. Optical signals within a waveguide may be controlled by an optical modulator, such as an optical phase shifter or the like.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0002] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0003] FIGS. **1-6** are cross-sectional views of intermediate stages in the manufacturing of waveguides, in accordance with some embodiments.

[0004] FIGS. **7A-7B** illustrate a bevel cleaning module and an associated bevel process.

DETAILED DESCRIPTION

[0005] The following disclosure provides many different embodiments, or examples, for implementing different features of the invention. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0006] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0007] According to various embodiments, waveguides are formed over a substrate. A waveguide includes a waveguide core as well as a surrounding portion of a cladding layer. A reshaping process is performed to alter a profile of the cladding layer, specifically, to form bevel profiles at the edges of the cladding layer. Forming the bevel profiles at the edges of the cladding layer may favorably affect cracking dynamics between the cladding layer and a waveguide core layer (for the waveguide cores). The risk of the cladding layer cracking may be reduced. On account of this reduced cracking risk, the cladding layer may be formed to a large thickness. Thus, the waveguide cores may have a large thickness, which may improve their optical performance.

[0008] FIGS. **1-6** are cross-sectional views of intermediate stages in the manufacturing of waveguides **503** (see FIG. **6**), in accordance with some embodiments. A waveguide **503** includes a portion of a waveguide cladding layer **101** and a waveguide core **501**. As subsequently described in

greater detail, a bevel profile of the waveguide cladding layer **101** will be reshaped before the waveguide cores **501** are formed. Specifically, the bevel profile of the waveguide cladding layer **101** is reshaped to reduce stress in the waveguide cladding layer **101**, which may decrease the risk of cracking when depositing a waveguide core layer (for the waveguide cores **501**). The manufacturing yield of the waveguides **503** may thus be increased. Reducing the risk of cracking allows the waveguide core layer to be deposited to a large thickness, which may improve the optical performance of the waveguides **503**.

[0009] In FIG. **1**, a substrate **100** is formed or provided. The substrate **100** may include a semiconductor substrate, such as a bulk semiconductor, an active layer of a semiconductor-on-insulator (SOI) substrate, or the like, which may be doped or undoped. The semiconductor substrate may include other semiconductor materials, such as germanium; a compound semiconductor including silicon carbide, gallium arsenic, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide; an alloy semiconductor including silicon germanium, gallium arsenide phosphide, aluminum indium arsenide, aluminum gallium arsenide, gallium indium arsenide, gallium indium phosphide, and/or gallium indium arsenide phosphide; or combinations thereof. Other substrates, such as multi-layered or gradient substrates, may also be used. The substrate **100** may be a wafer.

[0010] The semiconductor substrate has an active surface (e.g., the surface facing upward in FIG. **1**) and an inactive surface (e.g., the surface facing downward in FIG. **1**). The active surface is at a front-side of the substrate **100**, while the inactive surface is at a back-side of the substrate **100**. Devices are at the active surface of the semiconductor substrate. The devices may be active devices (e.g., transistors, diodes, etc.), capacitors, resistors, etc. The inactive surface may be free from devices. The device may be formed in a suitable front-end of line (FEOL) process.

[0011] An interconnect structure (not separately illustrated) may be over the active surface of the semiconductor substrate. The interconnect structure interconnects the devices of the semiconductor substrate to form an integrated circuit. The interconnect structure may be formed in a suitable back-end of line (BEOL) process. The interconnect structure may include one or more dielectric layer(s) and respective metallization pattern(s) in the dielectric layer(s). Acceptable dielectric materials for the dielectric layers include oxides such as silicon oxide, aluminum oxide, or the like; nitrides such as silicon nitride, silicon oxynitride; combinations thereof; or the like. The dielectric layer(s) may be formed of a low-k (LK) dielectric material such as carbon-doped silicon oxide, an extremely low-k (ELK) dielectric material such as porous carbon-doped silicon oxide, or the like. Other acceptable dielectric materials may be utilized. The metallization patterns may include conductive vias and/or conductive lines to interconnect the devices of the semiconductor substrate. The metallization patterns may be formed of a conductive material, such as a metal, such as copper, cobalt, aluminum, gold, combinations thereof, or the like. The metallization patterns may be formed by a damascene process, such as a single damascene process, a dual damascene process, or the like.

[0012] The substrate **100**, in the cross-sectional view, has an elongated oval shape. A center portion **100C** of the substrate **100** has a rectangular shape in the cross-sectional view, while edge portions **100E** of the substrate **100** have a rounded shape in the cross-sectional view. The substrate **100** may be circular in a top-down view (not separately illustrated), with the edge portions **100E** extending around each side of the center portion **100C** in the top-down view. Thus, the edge portions **100E** are adjacent either side of the center portion **100C** in the cross-sectional view.

[0013] A waveguide cladding layer **101** is formed over the substrate **100**. Portions of the waveguide cladding layer **101** will be components of waveguides. The waveguide cladding layer **101** may be formed of a dielectric material that is suitable for optics. In some embodiments, the waveguide cladding layer **101** is formed of an oxide dielectric material, such as silicon oxide, germanium oxide, combinations of these, or the like, which may be formed using a deposition method such as chemical vapor deposition (CVD), atomic layer deposition (ALD), or the like. However, any

suitable material and method of manufacture may be used to form the waveguide cladding layer **101**. The waveguide cladding layer **101** may be formed to any suitable thickness TH1. In some embodiments, the waveguide cladding layer **101** is formed to a thickness TH1 in the range of 2.0 μm to 4.0 μm (such as about 2.5 μm).

[0014] The edge portions **100E** of the substrate **100** each have a first length L1 in the cross-sectional view. In some embodiments, the first length L1 is in a range of 0.2 mm to 3.0 mm. If the first length L1 of the edge portions **100E** is less than 0.2 mm, then there may not be adequate space to form desired geometries (subsequently described) of the waveguide cladding layer **101** over the edge portions **100E** of the substrate **100**. If the first length L1 of the edge portions **100E** is greater than 3.0 mm, then the center portion **100C** of the substrate **100** may be too small to accommodate desired devices. The center portion **100C** of the substrate **100** has a second length L2 in the cross-sectional view. In some embodiments, the second length L2 is in a range of 280 mm to 299.6 mm.

[0015] In FIG. 2A, a profile of the waveguide cladding layer **101** (in the cross-sectional view) is reshaped by a bevel process **200**. The waveguide cladding layer **101** is removed from the outer edge of the substrate **100** by the bevel process **200**. As a result, the waveguide cladding layer **101** is divided into a front-side waveguide cladding layer **101F** (which remains on the front-side of the substrate **100**) and a back-side waveguide cladding layer **101B** (which remains on the back-side of the substrate **100**). The front-side waveguide cladding layer **101F** and the back-side waveguide cladding layer **101B** may be collectively referred to as the waveguide cladding layers **101**.

[0016] The bevel process **200** modifies the geometries of the waveguide cladding layers **101** where it overlaps the edge portions **100E** of the substrate **100**, to form bevel profiles **201** at the outer edges of the waveguide cladding layers **101**. The bevel process **200** may include an oxide clean process. The oxide clean process may include an etching process such as a wet etch, performed using a suitable cleaning solution. The etching process may be performed in multiple steps such that various portions of the waveguide cladding layers **101** are removed during each step of the etching process to form desired geometries for the waveguide cladding layers **101** overlapping the edge portions **100E** of the substrate **100**. The bevel process **200** will be subsequently described in greater detail for FIGS. 7A-7B.

[0017] The outer edges of the front-side waveguide cladding layer **101F** and the back-side waveguide cladding layer **101B** are reshaped by the bevel process **200**. After the bevel process **200**, the outer edges of the substrate **100** extend beyond the outer edges of the waveguide cladding layers **101**. The distance between an outer edge of the substrate **100** and a corresponding outer edge of a waveguide cladding layer **101** may be referred to as a “bevel depth” of that waveguide cladding layer **101**. In the illustrated embodiment, a bevel depth of the front-side waveguide cladding layer **101F** is substantially equal (within process variations) to a bevel depth of the back-side waveguide cladding layer **101B**. In another embodiment, a bevel depth of the front-side waveguide cladding layer **101F** is different than a bevel depth of the back-side waveguide cladding layer **101B**. For example, a bevel depth of the front-side waveguide cladding layer **101F** may be larger than a bevel depth of the back-side waveguide cladding layer **101B**. In some embodiments, the smallest bevel depth of the front-side waveguide cladding layer **101F** is about 1 nm, while the smallest bevel depth of the back-side waveguide cladding layer **101B** is about 0.2 mm.

[0018] FIG. 2B is a detailed view showing bevel profiles **201** of the waveguide cladding layers **101**, following the bevel process **200**. In this example, each bevel profile **201** is a staircase profile having multiple stairsteps **203**. For example, a staircase profile includes a first staircase **203A**, a second staircase **203B**, and a third staircase **203C**. Each staircase **203** may have its own bevel depth, with the bevel depths increasing in a direction extending away from the substrate **100**. In some embodiments, the first staircase **203A** has a first bevel depth BD1 in the range of 0.5 mm to 1.5 mm (such as about 1 mm), the second staircase **203B** has a second bevel depth BD2 in the range of 1.8 mm to 2.5 mm (such as about 2 mm), and the third staircase **203C** has a third bevel depth BD3 in the range of 2.8 mm to 5 mm (such as about 3 mm). If the bevel depths of the stairsteps **203** are too

great, then an insufficient quantity of stairsteps may be formed over the edge portions **100E** of the substrate **100** in the waveguide cladding layers **101**. If the bevel depths are too small then a distinct stairstep profile may not be achieved.

[0019] Although the stairstep profile illustrated in FIG. 2B depicts a three-stairstep profile, this is merely one example of the stairstep profile. The stairstep profile may have any number of stairsteps **203**. Additionally, the stairstep profile is merely one example of the bevel profile **201**, and any suitable bevel profile may be formed for the waveguide cladding layers **101**. For example, the bevel profiles **201** maybe sloped profiles, each of which has a bevel depth that increases continually in a direction extending away from the substrate **100**.

[0020] In FIG. 3, recesses **301** are formed in the front-side waveguide cladding layer **101F** by a suitable patterning process. The recesses **301** may be formed over only a front-side of the substrate **100**. The patterning process may include one or more photolithographic masking and etching processes.

[0021] The recesses **301** are each formed to a first depth **D1**. In some embodiments, the first depth **D1** is in the range of 100 nm to 1500 nm (such as about 350 nm or about 600 nm). Additionally, the recesses **301** are each formed to a first width **W1**. In some embodiments, the first width **W1** is in the range of 800 nm to 5000 nm. If the recesses **301** are formed to too large of a depth **D1** or too large of a width **W1**, then the risk of cracking between the front-side waveguide cladding layer **101F** and structures subsequently formed in the recesses **301** may be too great. If the recesses **301** are formed to too small of a depth or too small of a width, then the waveguide cores subsequently formed in the recesses **301** may be too small to adequately perform desired functions.

[0022] In FIG. 4, a waveguide core layer **401** is formed in the recesses **301** and over the waveguide cladding layers **101**. The waveguide core layer **401** extends along the bevel profiles **201** of the waveguide cladding layers **101**. The waveguide core layer **401** may be formed of a dielectric material that is suitable for optics. In some embodiments, the waveguide core layer **401** is formed of a nitride dielectric material, such as silicon nitride or the like, which may be formed using a deposition method such as chemical vapor deposition (CVD), atomic layer deposition (ALD), or the like. However, any suitable material and method of manufacture may be used to form the waveguide core layer **401**.

[0023] The material of the waveguide core layer **401** is different than the material of the waveguide cladding layers **101**. Due to the difference in refractive indices of the materials of the waveguide core layer **401** and the waveguide cladding layers **101**, the waveguide core layer **401** may have high internal reflections such that light is confined in the waveguide cores during operation, depending on the wavelength of the light and the reflective indices of the respective materials. In an embodiment, the refractive index of the material of the waveguide core layer **401** is higher than the refractive index of the material of the waveguide cladding layers **101**.

[0024] In some embodiments, the waveguide core layer **401** is formed of silicon nitride. The waveguide core layer **401** is formed to a large thickness **TH2**. In some embodiments, the waveguide core layer **401** is formed to a thickness **TH2** in the range of 150 nm to 3000 nm (such as about 350 nm or about 400 nm). Forming the waveguide core layer **401** of silicon nitride with a large thickness may allow the resulting waveguides to have higher optical confinement and/or lower optical losses. The optical performance of the resulting waveguides may thus be improved. In some embodiments, the material (e.g., silicon nitride) of the waveguide core layer **401** is formed by a low pressure chemical vapor deposition (LPCVD) process. The LPCVD process may be performed such that the waveguide core layer **401** conformally fills (and may overfill) the recesses **301**. In some embodiments, the LPCVD process is performed at a pressure in the range of 0.1 Pa to 0.4 Pa. Utilizing a LPCVD process may increase the thickness uniformity of the waveguide core layer **401**, even at large thicknesses.

[0025] The coefficient of thermal expansion (CTE) of the material of the waveguide core layer **401** may be mismatched with (e.g., significantly different than) the CTE of the material of the

waveguide cladding layers **101**. When the waveguide core layer **401** includes silicon nitride and the waveguide cladding layers **101** include silicon oxide, the CTE of the waveguide core layer **401** may be greater than the CTE of the waveguide cladding layers **101**. In some embodiments, the waveguide core layer **401** has a CTE of about 3.2 ppm/° C., the waveguide cladding layers **101** have a CTE of about 0.5 ppm/° C., and the substrate **100** has a CTE of about 2.6 ppm/° C. The CTE difference between the waveguide core layer **401** and the waveguide cladding layers **101** may be about 2.7 ppm/° C. Stress resulting from a CTE mismatch between the waveguide core layer **401** and the waveguide cladding layers **101** may be stored in the form of elastic energy. Forming the outer edges of the waveguide cladding layers **101** with the bevel profiles **201** helps reduce CTE-induced stress by varying local stress in the waveguide cladding layers **101**. Varying local stress in the waveguide cladding layers **101** may affect cracking dynamics, which may reduce the risk of the waveguide cladding layers **101** cracking, even at large thicknesses.

[0026] In FIG. 5, a removal process utilized to performed to removes excess portions of the waveguide core layer **401** from outside of the recesses **301** in the front-side waveguide cladding layer **101F**, thereby forming waveguide cores **501**. The waveguide cores **501** include the remain remaining portions of the waveguide core layer **401** in the recesses **301**. The removal process may be a planarization process, which may planarize a surface of the front-side waveguide cladding layer **101F** and the portions of the waveguide core layer **401** in the recesses **301**. In some embodiments, the planarization process is a chemical mechanical polish (CMP) process. After the planarization process, a top surface of the front-side waveguide cladding layer **101F** may be substantially coplanar (within process variations) with the top surfaces of the waveguide cores **501**. However, any suitable process, such as a mechanical process like a grinding process, may be utilized to planarize the material of the waveguide core layer **401** in the recesses **301** with the front-side waveguide cladding layer **101F**. After the removal process, the waveguide core layer **401** may remain on the back-side waveguide cladding layer **101B**, as well as on the outer edges of the substrate **100** and the front-side waveguide cladding layer **101F**. A top surface of the front-side waveguide cladding layer **101F** may be substantially coplanar (within process variations) with the portions of the waveguide core layer **401** remaining on the outer edges of the front-side waveguide cladding layer **101F**.

[0027] The resulting waveguides **503** include the waveguide cores **501** and portions of the front-side waveguide cladding layer **101F**. That is, a waveguide **503** includes a waveguide core **501**, as well as a surrounding portion of the front-side waveguide cladding layer **101F**. A surrounding portion of the front-side waveguide cladding layer **101F** may surround a waveguide core **501** on at least three sides in the cross-sectional view. The waveguide cores **501** have a thickness TH3 and a width W2. In some embodiments, the thickness TH3 is in the range of 600 nm to 800 nm, and the width W2 is in the range of 1000 nm to 2500 nm.

[0028] In FIG. 6, a back-side removal process **600** is performed to remove the waveguide core layer **401** and the back-side waveguide cladding layer **101B** from a back-side of the substrate **100**. As a result, the back-side of the substrate **100** is free of the waveguide cladding layers **101**. The waveguide core layer **401** may (or may not) also be removed from the outer edges of the substrate **100** and/or the outer edges of the front-side waveguide cladding layer **101F**. In some embodiments, the back-side removal process **600** is a wet clean or the like, performed using a suitable cleaning solution. The cleaning solution may include a solvent such as deionized (DI) water, an alkaline solution, an ammonia-based solution (including, e.g., Na, K, C, O, H, or the like), combinations thereof, or the like. The cleaning solution may be dispensed towards the back-side of the substrate **100** while the substrate **100** is rotated to distribute the cleaning solution across the back-side of the substrate **100**. The cleaning solution reacts with the waveguide core layer **401** and/or the waveguide cladding layers **101**. The cleaning solution used to remove the back-side waveguide cladding layer **101B** may be the same or different than the cleaning solution used to remove the waveguide core layer **401**. The cleaning solution may cover the back-side of the substrate **100**, and

may extend up along the outer edges of the substrate **100** during the back-side removal process **600**. Additionally, the cleaning solution may extend over the top-side of the substrate **100** by a small amount, such as a distance in the range of 0.8 mm to 2 mm, from the outer edges of the substrate **100**.

[0029] Following the back-side removal process **600**, the waveguides **503** remain at the front-side of the substrate **100**, with the waveguide core layer **401** and the back-side waveguide cladding layer **101B** having been removed from the back-side of the substrate **100**. Further, the front-side waveguide cladding layer **101F** remaining over the front-side of the substrate **100** has the bevel profiles **201** following the removal of the portions of the waveguide core layer **401** by the back-side removal process **600**.

[0030] The waveguides **503** may be used to interconnect the devices of the substrate **100**. For example, the substrate **100** may include photonic features. The photonic features transmit and receive optical signals, and convert optical signals to electrical signals. The devices of the substrate **100** may be interconnected by a signal path that includes electrical signals paths (that traverse conductive interconnects) as well as optical signal paths (that traverse the waveguides **503**), to form an integrated circuit.

[0031] Additional features (not separately illustrated) may be subsequently. For example, another waveguide cladding layer may be formed over the waveguide cores **501**. The waveguides **503** may further include surrounding portions of the overlying waveguide cladding layer. Additionally, die connectors may be formed for external connections to the integrated circuit of the substrate **100**. The die connectors may be adjacent to the waveguides **503**, and may extend through the front-side waveguide cladding layer **101F**. Likewise, passivation layer(s) may be formed to protect the substrate **100**.

[0032] The resulting wafer (including the substrate **100** and overlying waveguides **503**) may be singulated in subsequent processing to form integrated circuit dies. The singulation process singulates adjacent regions of the wafer from one another. The resulting integrated circuit dies are from the singulated regions of the wafer.

[0033] FIG. 7A shows a bevel cleaning module **700**, which may be utilized during the bevel process **200** (see FIG. 2A). The bevel cleaning module **700** is used for a wet clean process in this example, but other types of processes could be utilized to perform the bevel process **200** and achieve the bevel profiles **201**. The bevel cleaning module **700** includes a chuck **701**, a front-side nozzle **703**, a back-side nozzle **705**, an inlet **707**, and an outlet **709**. The substrate **100** is held by the chuck **701** within the bevel cleaning module **700**. The chuck **701** may be a vacuum chuck which utilizes reduced pressures in order to hold the substrate **100** in place. The chuck **701** holds the substrate **100** in place while the front-side nozzle **703** and the back-side nozzle **705** dispense a cleaning solution on the substrate **100**.

[0034] The front-side nozzle **703** may be positioned above the substrate **100** held by the chuck **701** and directed to dispense a cleaning solution from an interior of the substrate **100** towards the outer edges of the front-side of the substrate **100**. The back-side nozzle **705** may be positioned mirrored to the front-side nozzle **703** beneath the substrate **100** held by the chuck **701** and directed to dispense the cleaning solution from an interior of the substrate **100** towards the outer edges of the back-side of the substrate **100**. Multiple etch cycles may be performed, and the positions of the front-side nozzle **703** and the back-side nozzle **705** may be adjusted to change an interface of the cleaning solution with the substrate **100** between each etch cycle.

[0035] The inlet **707** is utilized to introduce the cleaning solution. The cleaning solution may include an etchant such as hydrofluoric (HF) acid or the like, suitable for removal of the waveguide cladding layers **101**. In some embodiments, the cleaning solution may also include a drying agent such as nitrogen (N.sub.2). However, any suitable cleaning solution may be utilized in the removal of the waveguide cladding layers **101** during the bevel process **200**. Further, the outlet **709** may act as a drain where removed portions of the waveguide cladding layers **101** and the cleaning solution

may be extracted from the bevel cleaning module **700**. The outlet **709** may include an exhaust for process gases.

[0036] FIG. 7B shows a simplified process view of the bevel process **200** in the bevel cleaning module **700**, in accordance with some embodiments. The chuck **701** holds the substrate **100** while the front-side nozzle **703** and the back-side nozzle **705** dispense a cleaning solution **750** towards the outer edges of the substrate **100** to form bevel profiles **201** (see FIG. 2A) in the outer edges of the waveguide cladding layers **101**. In some embodiments, the bevel process **200** removes portions of the waveguide cladding layers **101** in a series of etch cycles, in which the cleaning solution **750** is dispensed in desired locations. The positions of the front-side nozzle **703** and the back-side nozzle **705** may be adjusted between each of the etch cycles so as to interface with the waveguide cladding layers **101** at different positions, thereby reshaping the outer edges of the waveguide cladding layers **101** into desired bevel profiles **201**. For example, the front-side nozzle **703** and the back-side nozzle **705** may move further away from the outer edge of the substrate **100** after each etch cycle, or may move closer to the outer edge of the substrate **100** after each etch cycle. In some embodiments, during the bevel process **200**, the substrate **100** is held to a process temperature in the range of about 50° C. to about 80° C. In some embodiments where the bevel profiles **201** are stairstep profiles, an etching depth for each stairstep **203** (e.g., a bevel etching depth) may be in a range of 0.2 mm to 3.0 mm.

[0037] Embodiments may achieve advantages. By forming the bevel profiles **201** in the front-side waveguide cladding layer **101F**, local stress at the edge portions **100E** of the substrate **100** may be reduced, which may mitigate the risk of cracks forming during deposition of the waveguide core layer **401** with a large thickness. As a result, the waveguide core layer **401** may be deposited to a large thickness in a single deposition process, simplifying the manufacturing process and reducing processing costs associated with multi-step deposition processes. The optical performance and yield of the waveguides **503** may thus be improved. Specifically, the waveguide cores **501** having a large thickness may allow the resulting waveguides **503** to have higher optical confinement and/or lower optical losses.

[0038] In an embodiment, a method includes: depositing a waveguide cladding layer over a substrate, the substrate having a center portion in a cross-sectional view and an edge portion adjacent the center portion in the cross-sectional view; reshaping the waveguide cladding layer to form a bevel profile in the waveguide cladding layer over the edge portion of the substrate; forming a recess in the waveguide cladding layer; depositing a waveguide core layer in the recess and over the waveguide cladding layer, the waveguide core layer extending along the bevel profile of the waveguide cladding layer; and planarizing the waveguide core layer and the waveguide cladding layer to form a waveguide core, the waveguide core including a portion of the waveguide core layer in the recess. In some embodiments of the method, the waveguide core layer is deposited by a low pressure chemical vapor deposition process. In some embodiments of the method, the waveguide cladding layer is formed of silicon oxide and the waveguide core layer is formed of silicon nitride. In some embodiments of the method, during the reshaping of the waveguide cladding layer, the substrate is maintained at a temperature between 50° C. to 80° C. In some embodiments of the method, reshaping the waveguide cladding layer includes performing a series of etch cycles using an etchant, an interface between the etchant and the waveguide cladding layer being changed between each of the etch cycles. In some embodiments of the method, the bevel profile is a stairstep profile. In some embodiments of the method, the waveguide core layer has a thickness in a range of 150 nm to 3000 nm.

[0039] In an embodiment, a method includes: forming a front-side waveguide cladding layer over a substrate, an outer edge of the front-side waveguide cladding layer having a first bevel profile, the front-side waveguide cladding layer including a first material; forming a recess in the front-side waveguide cladding layer; and forming a waveguide core in the recess, a top surface of the waveguide core being substantially coplanar with a top surface of the front-side waveguide

cladding layer, the waveguide core including a second material, the second material having a higher refractive index than the first material. In some embodiments of the method, the first bevel profile is a staircase profile including a plurality of stairsteps, each of the stairsteps having a different bevel depth. In some embodiments, the method further includes: forming a back-side waveguide cladding layer over the substrate, an outer edge of the back-side waveguide cladding layer having a second bevel profile. In some embodiments of the method, the front-side waveguide cladding layer has a first bevel depth measured from the outer edge of the substrate, the back-side waveguide cladding layer has a second bevel depth measured from the outer edge of the substrate, and the first bevel depth is substantially equal to the second bevel depth. In some embodiments of the method, the front-side waveguide cladding layer has a first bevel depth measured from the outer edge of the substrate, the back-side waveguide cladding layer has a second bevel depth measured from the outer edge of the substrate, and the first bevel depth is different than the second bevel depth. In some embodiments, the method further includes: removing the back-side waveguide cladding layer from the substrate while the front-side waveguide cladding layer remains over the substrate.

[0040] In an embodiment, a device includes: a substrate having an edge portion with a rounded shape in a cross-sectional view; a waveguide cladding layer over a front-side of the substrate, an outer edge of the waveguide cladding layer having a bevel profile, the bevel profile overlapping the edge portion of the substrate with the rounded shape; and a waveguide core in the waveguide cladding layer, a top surface of the waveguide core being substantially coplanar with a top surface of the waveguide cladding layer. In some embodiments of the device, the bevel profile is a staircase profile. In some embodiments of the device, the staircase profile includes stairsteps, each of the stairsteps having a different bevel depth measured from an outer edge of the substrate. In some embodiments of the device, the waveguide core has a thickness in a range of 600 nm to 800 nm. In some embodiments of the device, a back-side of the substrate is free of waveguide cladding layers. In some embodiments of the device, the waveguide cladding layer includes a first material, the waveguide core includes a second material, and the second material has a higher refractive index than the first material. In some embodiments of the device, the first material is silicon oxide and the second material is silicon nitride.

[0041] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A method comprising: depositing a waveguide cladding layer over a substrate, the substrate having a center portion in a cross-sectional view and an edge portion adjacent the center portion in the cross-sectional view; reshaping the waveguide cladding layer to form a bevel profile in the waveguide cladding layer over the edge portion of the substrate; forming a recess in the waveguide cladding layer; depositing a waveguide core layer in the recess and over the waveguide cladding layer, the waveguide core layer extending along the bevel profile of the waveguide cladding layer; and planarizing the waveguide core layer and the waveguide cladding layer to form a waveguide core, the waveguide core comprising a portion of the waveguide core layer in the recess.
2. The method of claim 1, wherein the waveguide core layer is deposited by a low pressure chemical vapor deposition process.

3. The method of claim 1, wherein the waveguide cladding layer is formed of silicon oxide and the waveguide core layer is formed of silicon nitride.
4. The method of claim 1, wherein during the reshaping of the waveguide cladding layer, the substrate is maintained at a temperature between 50° C. to 80° C.
5. The method of claim 1, wherein reshaping the waveguide cladding layer comprises performing a series of etch cycles using an etchant, an interface between the etchant and the waveguide cladding layer being changed between each of the etch cycles.
6. The method of claim 1, wherein the bevel profile is a staircase profile.
7. The method of claim 1, wherein the waveguide core layer has a thickness in a range of 150 nm to 3000 nm.
8. A method comprising: forming a front-side waveguide cladding layer over a substrate, an outer edge of the front-side waveguide cladding layer having a first bevel profile, the front-side waveguide cladding layer comprising a first material; forming a recess in the front-side waveguide cladding layer; and forming a waveguide core in the recess, a top surface of the waveguide core being substantially coplanar with a top surface of the front-side waveguide cladding layer, the waveguide core comprising a second material, the second material having a higher refractive index than the first material.
9. The method of claim 8, wherein the first bevel profile is a staircase profile comprising a plurality of stairsteps, each of the stairsteps having a different bevel depth.
10. The method of claim 8, further comprising: forming a back-side waveguide cladding layer over the substrate, an outer edge of the back-side waveguide cladding layer having a second bevel profile.
11. The method of claim 10, wherein the front-side waveguide cladding layer has a first bevel depth measured from the outer edge of the substrate, the back-side waveguide cladding layer has a second bevel depth measured from the outer edge of the substrate, and the first bevel depth is substantially equal to the second bevel depth.
12. The method of claim 10, wherein the front-side waveguide cladding layer has a first bevel depth measured from the outer edge of the substrate, the back-side waveguide cladding layer has a second bevel depth measured from the outer edge of the substrate, and the first bevel depth is different than the second bevel depth.
13. The method of claim 10, further comprising: removing the back-side waveguide cladding layer from the substrate while the front-side waveguide cladding layer remains over the substrate.
14. A device comprising: a substrate having an edge portion with a rounded shape in a cross-sectional view; a waveguide cladding layer over a front-side of the substrate, an outer edge of the waveguide cladding layer having a bevel profile, the bevel profile overlapping the edge portion of the substrate with the rounded shape; and a waveguide core in the waveguide cladding layer, a top surface of the waveguide core being substantially coplanar with a top surface of the waveguide cladding layer.
15. The device of claim 14, wherein the bevel profile is a staircase profile.
16. The device of claim 15, wherein the staircase profile comprises stairsteps, each of the stairsteps having a different bevel depth measured from an outer edge of the substrate.
17. The device of claim 14, wherein the waveguide core has a thickness in a range of 600 nm to 800 nm.
18. The device of claim 14, wherein a back-side of the substrate is free of waveguide cladding layers.
19. The device of claim 14, wherein the waveguide cladding layer comprises a first material, the waveguide core comprises a second material, and the second material has a higher refractive index than the first material.
20. The device of claim 19, wherein the first material is silicon oxide and the second material is silicon nitride.

